Date: 24/3

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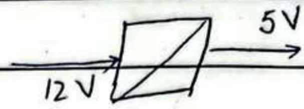
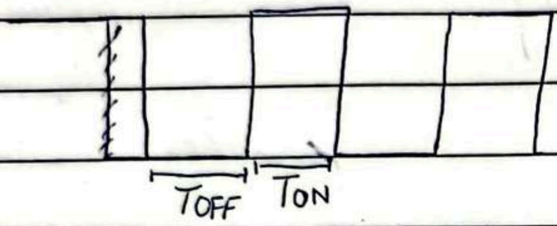
CH # 7 SWITCHING POWER SUPPLIES (DC to DC)

CASE-I:

- Vin → Constant] fixed , & also D - fixed
  Vout → Constant]
  ↓5V

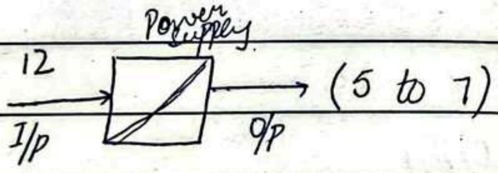




CASE-II:

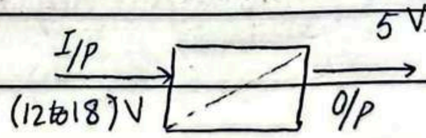
- Vin → Constant/fixed
  Vout → Variable

⇒ D → Variable



CASE-III:

- Vin → variable
  Vout → fixed
  D → variable



CASE-IV

- Vin → variable
  Vout → variable

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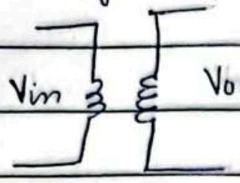
BUCK REGULATORS / CONVERTERS ($V_{in} > V_o$)

- Input is high, o/p. is low

eg#

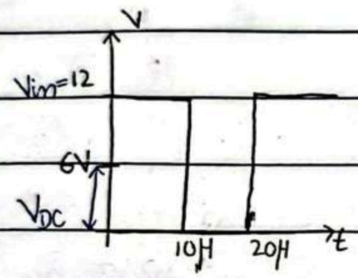


Step down transformer

★ $V_{out} = D \times V_{in}$

- ideally $D = 1$

1)



$$V_{avg} = 12 \times \frac{10}{20} = 6V$$

TON
TON+TOFF

2) $V_{avg} = 12 \times (5/20) = 3V$

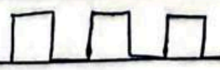
3) $V_{in} = 7V$

$V_{out} = 6V$

$$D = \frac{V_{out}}{V_{in}} = \frac{6}{7} = 0.85$$

- Switch driver is used to control switch.
- Schotkky diode in parallel with switch - used to prevent switch.
(zener)
- O/p of switch is rectangle, filter required.

26/3 PWM



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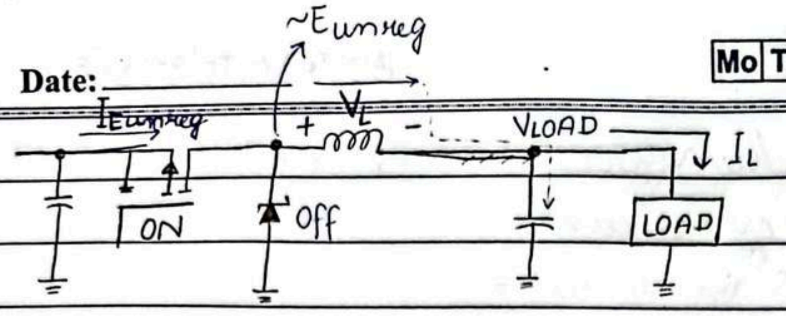
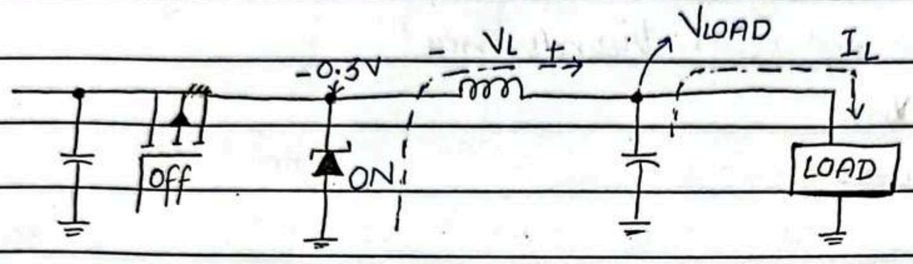


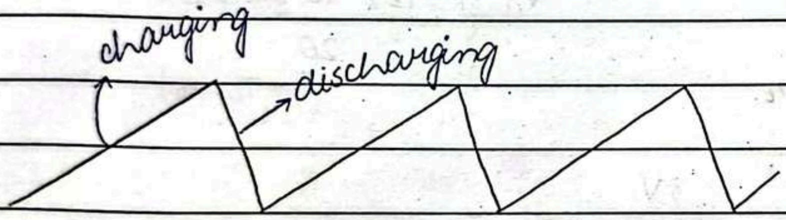
Change part of the cycle

Buck regulator operation

* VL & VLOAD are different



Discharge part of the cycle



O/P graph

*[current rating $> 1.3\, I_{DC,LOAD}$
PIV rating $> 1.3\, E_{unreg,max}$

$$V_L = L \frac{\Delta i}{\Delta t}$$

* inductor - opposes change in current

$$V_L = E_{unreg} - V_{LOAD}$$
$$= V_i - V_o$$
$$= V_i - DV_i$$
$$= (1-D)V_i$$

*[Analysis → calculate Δi
Design → calculate L

Δi - AC current capacitor will entertain.

*ripple current - depends on inductor

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$\Delta t = t_{ON} = DT$ (duty cycle is set by the relationship of i/p to o/p $\Rightarrow D = V_{LOAD} / E_{unreg}$)

* Select $\Delta i = 0.25\, I_{DC}$ to $0.7\, I_{DC}$ (Ripple current)

→* Load current will be smooth current.

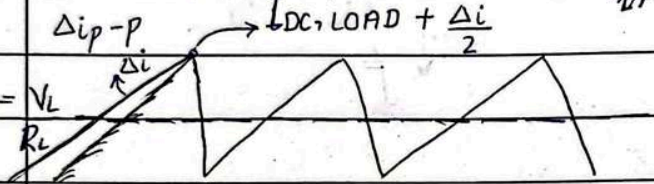
* → At or below $200m\,A_{DC}$, allow longer ripple current.

→ Above $1A_{DC}$, restrict ripple current to 25%.

→ In analysis, ripple current is set by inductor.

* Inductor max current $= i_{P_{inductor}} = I_{DC,LOAD} + \dfrac{\Delta i}{2}$

$I_{DC,LOAD} = \dfrac{V_L}{R_L}$



(continuous)

EXAMPLE # 7.2

$V_{in} = E_{unreg} = 12V$, $f = 100kHz \Rightarrow T = 1/100k = 0.01ms$

$V_{LOAD} = V_o = 5V$, $L = 220\,\mu H$

$R_L = 10\,\Omega$

i) Duty cycle $= D = ?$, ii) on time $= \Delta t = ?$

iii) ripple current $= \Delta i = ?$, iv) avg. Inductor Current $= I_{DC,LOAD} = ?$

v) Peak inductor current $= I_{L,Peak} = ?$

voltage across inductor

$V_L = E_{unreg} - V_{LOAD}$

$= 12 - 5 = 7V$

i) $D = \dfrac{V_{LOAD}}{E_{unreg}} = \dfrac{V_o}{V_{in}} = \dfrac{5}{12} = 42\%$

ii) $\Delta t = D \times T = 0.42\,(0.01m)$

$= 4.2\,\mu s$

→

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POWER ELECTRONICS

CHAPTER # 7

EXAMPLE # 7.2 (Remaining)

iii) $\Delta i = ?$

$$V_L = L \frac{\Delta i}{\Delta t}$$

$$\Delta i = \frac{V_L \Delta t}{L} = \frac{7(4.2\mu)}{220\mu}$$

$\Delta i = 134 mA_p$

$V_L = 12 - 5$ OR $V_L = (1-D)V_{in}$

$V_L = 7V$ $\Rightarrow (1-D)V_{in} = L(\Delta i/\Delta t)$

$$\Delta i = \frac{(1-0.42)(12)(4.2\mu)}{220\mu}$$

$\Delta i = 133 mA$

iv) $I_{DC,LOAD} = \frac{V_{LOAD}}{R_L} = \frac{5 V_{DC}}{10}$

$= 500 mADC$

* To verify:

$\Delta i / I_{DC,LOAD} = \frac{134 mApp}{500 mADC}$

$= 0.27 \Rightarrow 27\%$

$\Rightarrow$ close to 25% $\rightarrow$ (reasonable)

v) $I_{L,peak} = I_{DC,LOAD} + \frac{\Delta i}{2}$

$= 500 m ADC + \frac{134 m App}{2}$

$= 567 mA$

OR using $\Delta i = 133 mA$

$\Rightarrow I_{L,Peak} = 500m + \frac{133m}{2}$

$= 565 mA$

7.1 * BUCK REGULATORS: (O/P smaller than I/P) (Remaining)

→ A key parameter ~~of~~ for Cin is its equivalent series resistance.

→ Large capacitors, Cin & Cout may have hundred milliohms of internal resistance.

→ Current flowing off and onto these capacitors produces a voltage drop & forces capacitor to dissipate power (producing heat).

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* • $I_{rms}\ C_{in} > 0.5\ I_{DC,LOAD}$ (I/p capacitor ratings)
• $V_{Cin} > 1.5\ E_{unreg,max}$

→ * Any capacitor above 100µF should work.

* • $\Delta V_{PP,LOAD} = \Delta i_{pp,inductor} \times ESR_{Cout}$ (O/p capacitor's specifications)

* • $I_{RMS,Cout} = \dfrac{\Delta i_{pp,inductor}}{\sqrt{3}}$

* • $C \approx 500\ \dfrac{\mu F}{A} \times I_{DC,LOAD}$

EXAMPLE # 7.3

$I_{DC,LOAD} = 500\,mADC$, $\Delta i_{pp} = 134\,mApp$,

$D = 0.42$

i) $I_{RMS} = ?$ (using trapezoid equation)

ii) $I_{RMS} = ?$ (using simple rectangle with $I_p = I_{DC,LOAD}$)

i) For trapezoid:

$$I_{RMS} = \sqrt{\frac{D}{3}\left(I_p^2 + I_p I_{min} + I_{min}^2\right)} \quad (i)$$

where,

$$I_p = I_{DC} + \frac{\Delta i_{pp}}{2} \quad \& \quad I_{min} = I_{DC} - \frac{\Delta i_{pp}}{2}$$

$$= 500m + \frac{134m}{2} \quad \& \quad = 500m - \frac{134m}{2}$$

$$I_p = 567\,mAp \quad \& \quad I_{min} = 433\,mAp$$

⇒ (i) $I_{RMS} = \sqrt{\dfrac{0.42}{3}\left[(567m)^2 + (567m)(433m) + (433m)^2\right]}$

$= 325\,mArms$

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ii) For rectangle (with flat top of 500mAp)

$I_{RMS} = \sqrt{D} \times I_p = \sqrt{0.42} \times 500m\,Ap$

$= 324m\,A_{RMS}$

$\Delta i = \frac{\Delta i_{pp}}{I_{DC,LOAD}} \times 100 = \frac{134}{500} \times 100$

$\Delta i = 26.8\% = 27\%$

7/4 EXAMPLE #

$I_{DC,LOAD} = 500mA$, $\Delta i_{pp} = 134mA$,

$D = 0.42$ ($R_{ON} = 0.6\Omega$ at 150°C)

i) Power dissipation of IRF9530 = ?

$P_Q = \overset{D}{0.42} \overset{\times}{\times} \overset{I_{DC,LOAD}}{(500m)^2} \overset{\times R_{ON}}{\times 0.6}$

$= 63mV$

ii) Power dissipation by Schottky diode = ?

$P_{DIODE} = D \times V_{ON} \times I_{DC,LOAD}$

$= 0.42 \times 0.3 \times 500m$

$= 63\,mW$

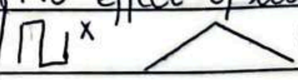
iii) $P_L = ?$, $R_L = 10\Omega$

$P_L = I^2_{DC,LOAD} \times R_L$

$= (500m)^2 (10)$

$= 2.5W$

[NO effect of load)



(continuous)]

iv) efficiency $= \eta = ?$

$\eta = \frac{P_L}{P_T} \times 100 = \frac{2.5}{63m \times 63m \times 2.5} \times 100$

when I/p is known

$\Rightarrow \eta = \frac{\text{input} - \text{losses}}{\text{input}}$

$\eta = 95.2\%$

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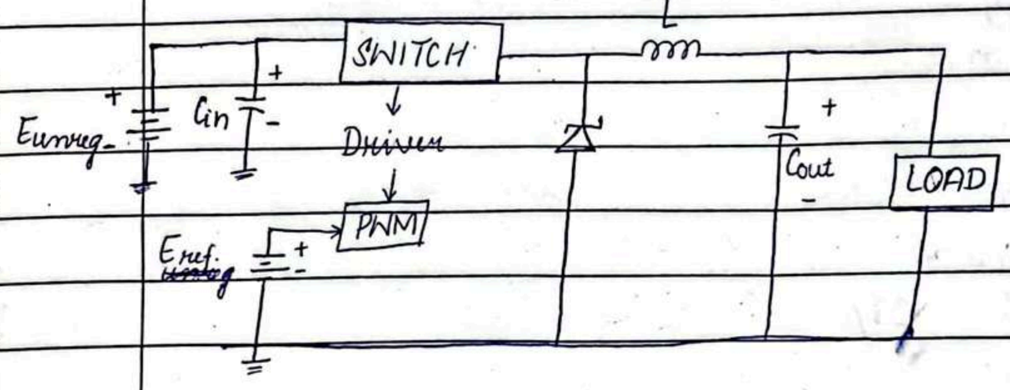
BUCK REGULATOR (LM 2959)



- $P_Q = D \times I_{DC,LOAD} \times V_{sat,max}$
- $P_{IC} = I_{quise} \times E_{unreg}$
- Total Power dissipation $= P_Q + P_{IC}$

EXAMPLE #

convert unregulated input voltage of 12V to 5 V_{DC}.

$R = 10\Omega$, $T_A = 80^\circ C$, $f = 150kHz$, $V_{ref} = 1.23V$

Sol. Designing Buck regulator.

LM2595 specifications

$V_{sat\,(max)} = 1V_p$

$\theta_{JC} = 2^\circ C/W$

$\theta_{JA\,(bare)} = 50^\circ C/W$

$L = ?$

$$V_L = L\frac{\Delta i}{\Delta t}$$

$$V_L = 12 - 5 = 7V$$

$$I_{DC,LOAD} = \frac{5}{10} = 0.5A = 500mA.$$

Select $\Delta i = 0.25$ to $0.7\, I_{DC}$

$$\Delta i_{pp} = 0.25 \times 500m$$
$$= 125m$$

$$T = \frac{1}{f} = \frac{1}{150k} = 6.7\mu s$$

$$\Delta t = D \times T$$
$$= 0.42 \times 6.7\mu s$$
$$= 2.8\mu s$$

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$V_o = DV_{in} \Rightarrow \frac{5}{12} = D$

$\Rightarrow L = 157\mu H \approx 220\mu H$ $\qquad (V_L = L\,\overset{125m}{\Delta i}/\Delta t)$

$I_{DC,LOAD} = 500mA$

$I_p = \frac{500 + 125}{2} = 563mA$

* $C_{out} = 250\mu F$

$C_{in} > 100\mu F$

$I_{C_{in}} > 0.5 \times I_{DC,LOAD} = 250\ mA$

$V_{C_{in}} \geq 1.5\, E_{unreg} = 1.5 \times 12 = 18V$

$C_{out} \approx 500\mu F/A \times I_{DC,LOAD}$

$P_T = P_Q + P_{IC}$

$= (10m \times 12) + (0.42 \times 500m \times 1)$

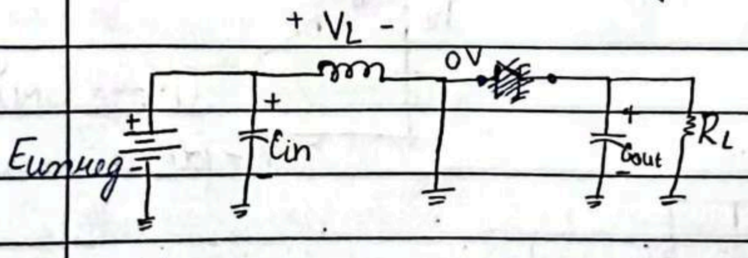
$= 0.33W$

$\theta_{SA} = \frac{150-80}{0.33} - 2 - 0.2$

$= 210C/W$

* 14/4 BOOST REGULATORS / CONVERTERS

- Low side instead of high side.
- transistor OFF, diode ON and vice versa.
- Transistor ON: $V_L = E_{unreg} = L\frac{\Delta i}{\Delta t}$ (when charging).



when transistor is ON/ diode is OFF.

$V_L = E_{unreg} - 0$

$V_L = E_{unreg} = L\frac{\Delta i}{\Delta t}$

current – [inductor – ramp
capacitor – constant.

★ Δi_{pp} – [for buck $= 0.25 \times I_{DC,LOAD}$
for boost $= 0.25 \times I_{DC,I/P}$

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- Boost regulators are 90% efficient.
- ★ $P_{LOAD} = P_{in} \times 0.9$ OR $V_{DC,LOAD} \times I_{DC,LOAD} = 0.9\, E_{unreg} \times I_{DC,I/P}$ → (i)

→ When transistor is ON:

ANALYSIS:

$$\Delta i = \frac{E_{unreg}\, \Delta t}{L}$$

DESIGN:

- $\Delta i = 25\%$ of $I_{DC,\cancel{LOAD}\ I/P}$
- $L = \dfrac{E_{unreg}\, \Delta t}{\Delta i}$
- $\Delta t = D \times T$
- $I_{L,PEAK} = \cancel{I_L}\ I_{DC,I/P} + \dfrac{\Delta i}{2}$ ∴ L – inductor

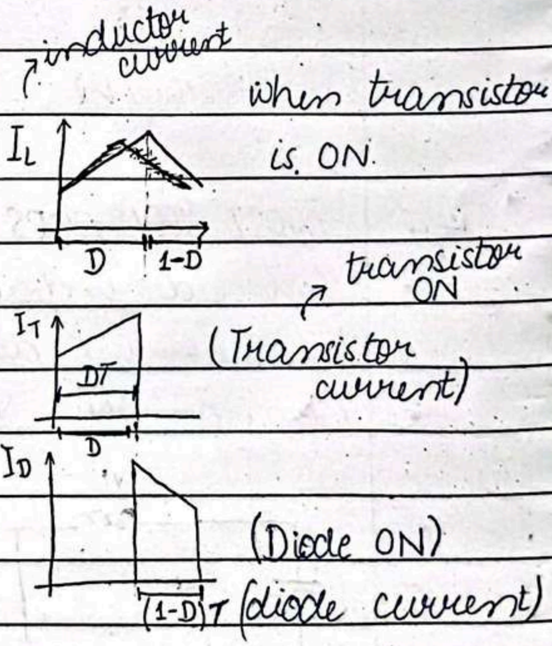
  inductor
- $V_{LOAD} = E_{unreg} + V_L$

⇒ (i) $I_{DC,I/P} = \dfrac{V_{DC,LOAD} \times I_{DC,LOAD}}{0.9\, E_{unreg}}$

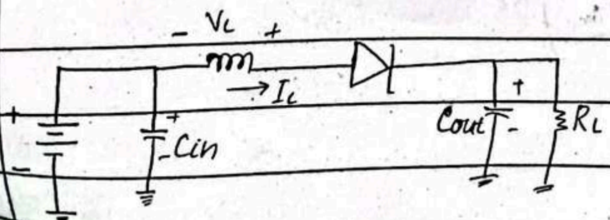
- $I_L = \dfrac{V_{DC,LOAD} \times I_{DC,LOAD}}{0.9\, E_{unreg}}$

⇒★ $I_{DC,I/P} = I_L$

- $I_{avg,L} = \dfrac{V_{DC,LOAD} \times I_{DC,LOAD}}{0.9\, E_{unreg}}$



4. $I_{DC} = D \times I_L$



(when transistor – OFF, diode ON)

★ ⇒ $I_{DC,LOAD} = C_{out} \times \dfrac{\Delta V_{LOAD}}{\Delta t}$

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- $I_{avg,Q} = D \times I_{DC, I/P}$ } transistor
- $I_{RMS,Q} = \sqrt{D} \times I_{DC, I/P}$

- $I_{avg,D} = (1-D) \times I_{DC, I/P}$ (D → diode)
- ~~Iavg, D~~ $I_{RMS,D} = \sqrt{(1-D)} \times I_{DC, I/P}$

- O/p higher than I/p (always)
- when transistor is OFF:

$-E_{unreg} - V_L + V_{LOAD} = 0$

* $V_{LOAD} = V_L + E_{unreg}$

*• Keep $\Delta V_{LOAD} < 20\,mV_{PP}$ (if ΔV_{PP} is not given)

*• $V_L = \dfrac{L\,\Delta i}{(1-D)T}$ (i) (discharging)

• $E_{unreg} = \dfrac{L\,\Delta i}{DT} \Rightarrow L = E_{unreg} \times \dfrac{DT}{\Delta i}$ (ii)

$\Rightarrow$ (ii) in (i)

$$V_L = E_{unreg} \times \frac{D\cancel{T}}{\cancel{\Delta i}} \times \frac{\cancel{\Delta i}}{(1-D)\cancel{T}}$$

$$V_L = E_{unreg} \times \frac{D}{(1-D)}$$

$$V_{LOAD} = E_{unreg} + V_L$$

$$= E_{unreg} + E_{unreg}\left(\frac{D}{1-D}\right)$$

*• $V_{LOAD} = E_{unreg}\left(1 + \dfrac{D}{1-D}\right) = E_{unreg}\left(\dfrac{1}{1-D}\right)$ ∴ $D < 1$

$$\downarrow$$

$$\frac{1-\cancel{D}+\cancel{D}}{1-D}$$

$D - \left[\begin{array}{l} \text{for buck regulator} = V_o/V_{in} \\ \text{for boost regulator} = 1 - \frac{V_{in}}{V_o} \end{array} \right.$

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- $C_{OUT} = \frac{I_{DC,LOAD} \times \Delta t}{\Delta V_{LOAD}} = \frac{I_{DC,LOAD} \times DT}{\Delta V_{LOAD}}$.
- $\Delta V_{pp} = \Delta i_{pp,inductor} \times ESR_{Cout}$, ~~Cout = 1500 μF / A × I_{DC,LOAD}~~

EXAMPLE # 7.6 (Boost regulator → O/P > I/P)

$V_{in} = 12 V_{DC}$, $V_{LOAD} = 22 V_{DC}$, $0.7 A_{DC} = I_{LOAD}$

$f = 100kHz \Rightarrow T = 1/100k = 1 \times 10^{-5}$

$R_{ON} = 0.4 \Omega$, diode drops $= 0.5V$

i) $P_{LOAD} = P_{in} \times 0.9$

$= (12 \times 1.425) \times 0.9$ $\qquad I_{DC,I/P} = \frac{22 \times 0.7}{0.9 \times 12} = 1.425$

$P_{LOAD} = 15.39 W \approx 15.4 W$

ii) $I_{DC,I/P} = 1.425A$

iii) $\Delta i_{pp} = 0.\cancel{175}\,35$ (25% of $I_{DC,I/P}$) $\qquad \therefore 0.25 \times 1.4 = 0.35$

$\Delta t = DT$

$L = \frac{E_{unreg} \times \Delta t}{\Delta i_{pp}} = \frac{12 \times (0.45)(1 \times 10^{-5})}{\cancel{0.175}\ 0.35}$ $\qquad {}^*D = 1 - \frac{E_{unreg}}{V_{LOAD}} = 0.45$

$L = \cancel{3.08 \times 10^{-4}}\ 1.54 \times 10^{-4} = 154 \mu H$

iv) Current rating $= I_{L,P} = 1.4 + \frac{0.35}{2}$

$= 1.6 A_p$

v) $P_Q = D \times I_L^2 \times R_{ON}$

$= 0.45 \times 1.4^2 \times 0.4$

$= 353 mW$

* Buck → I/p > O/p
Boost → I/p < O/p

* for Cout, use 20m if ΔV_{pp} is not given

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vi) * $P_{DIODE} = (1-D) \times I_L \times V_D = (1-0.45) \times 1.4 \times 0.5$
$= 390\ mW$

vii) $C_{out} = \frac{0.7 \times 0.45 \times 1 \times 10^{-5}}{20m} \rightarrow {}^{*}V_L < 20m$
$= 1.575 \times 10^{-4} = 158\mu F$

16/8 BOOST REGULATOR LM2585 (O/p > I/p)

- $V_{DC,LOAD} = 1.23 V_{DC} \left(1 + \frac{R_f}{R_i}\right)$, $1k\Omega < R_i < 5k\Omega$
- $P_Q = I_{rms,Q}^2 \times R_{ON,Q}$: $I_{rms,Q} = \sqrt{D}\ I_{DC,LOAD}$, ${}^{*}R_{ON,Q} = 0.15$
- $P_{IC} = \frac{1}{50} \times D \times I_{DC,I/p} \times E_{unreg}$
- Total power $= P_Q + P_{IC}$

EXAMPLE # 7.7
$E_{unreg} = 12 V_{DC}$, $V_{LOAD} = V_o = 22 V_{DC}$, $I_{LOAD} = 0.7 A_{DC}$
$T_A = 80°C$

i) $V_{DC,LOAD} = (1.23 V_{DC}) \left(1 + \frac{R_f}{R_i}\right)$

$\frac{R_f}{R_i} = \frac{22}{1.23 V_{DC}} - 1$

$\frac{R_f}{R_i} = 16.9$ (i)

Assume, $R_i = 1.3k\Omega$
$\Rightarrow$ (i) $R_f = 16.9 \times 1.3k = 21.9k\Omega \approx 22k\Omega$

EXAMPLE #2 (PRACTICE PROBLEM)

BOOST REGULATOR

$E_{unreg} = 22 V_{DC}$, $I_{DC, I/P} = 0.7 A_{DC}$,

$V_o = V_{LOAD} = 56 V_{DC}$

$f = 100 kHz \Rightarrow T = 1 \times 10^{-5} s$

$R_{ON} = 0.4 \Omega$, $V_D = 0.5 V$

i) $P_{LOAD} = P_{in} \times 0.9 = V_{in} \times I_{DC, I/P} \times 0.9$

$= 22 \times 0.7 \times 0.9$

$= 13.86 W$

ii) $\Delta i_{pp} = 0.25 \times 0.7$

$= 0.175 A$

$\Delta t = DT$

$= 6.07 \mu s$

$\because D = 1 - \frac{V_{in}}{V_{out}}$

$= 0.607$

$L = \frac{V_{in} \times \Delta t}{\Delta i_{pp}} = \frac{22 \times 6.07 \mu}{0.175}$

$L = 763 \mu H$

Current ratings = $I_{L,PEAK}$

$$= I_{I/P} + \frac{\Delta i_{PP}}{2}$$

$$= 0.7 + \frac{0.175}{2} = 0.7875$$

$x, I_{I/P} = I_L$

$Q = D \times I_L^2 \times R_{ON} = 0.607 \times (0.\cancel{7})^2 \times 0.4$

$= 119 mW$

$P_D = (1-D) \times I_L \times V_D$

$= (1-0.607) \times (0.7)(0.5)$

$= 137.5 mW$

$$C_{out} = \frac{I_{DC,LOAD} \times \Delta t}{\Delta V_{LOAD}} \quad (i)$$

$$I_{DC,LOAD} = \frac{P_{LOAD}}{V_{LOAD}} = \frac{13.86}{56} = 0.2475$$

$$C_{out} = \frac{0.2475 \times 6.07 \mu}{20 m}$$

$= 75.1 \mu F$

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i) $P_{in} = (12 \times 1.425)$ $\quad \because I_{DC, I/P} = 1.425A$

$= 17.1W$

$= (\sqrt{D}\ I_{DC,I/P})^2 \times R_{ON,Q}$

ii) $P_Q = (\sqrt{0.45}(1.4))^2 (0.15)$

$= 0.1323W \quad = 132\,mW$

iii) $P_{IC} = \frac{1}{50} \times 0.45 \times 1.4 \times 12$

$= 0.1512 \quad = 151\,mW$

iv) $P_T = 132m + 151m$

$= 283\,mW$

v) $T_J = 80 + (0.283)(\overbrace{65^\circ}^{\theta_{JA}})$

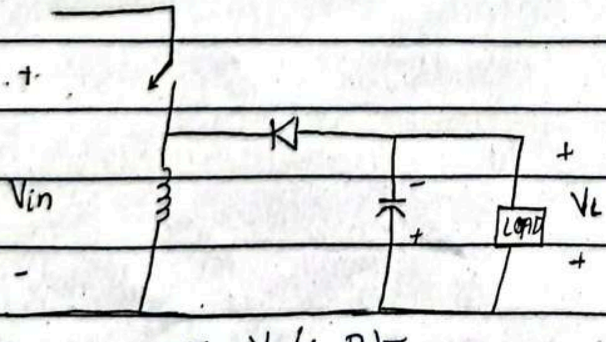
$= 98.4^\circ C \quad < 110^\circ C$ (no heat sink)

$V_o = D \times V_{in}$ $\quad V_o = \frac{1}{1-D} \times V_{in}$

BUCK-BOOST CONVERTER:

$$V_o = \frac{D}{1-D} \times V_{in}$$

* $D > 0.5$ (50%) (Boost converter)
* $D < 0.5$ (Buck converter)



$V_L = V_{in} DT - V_o(1-D)T$

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EXAMPLE #4 (PROBLEM #7.3) (Buck regulator)

$E_{unreg} = 9V_{DC}$, $V_{LOAD} = 5V_{DC}$, $I_{DC,\,LOAD} = 0.8A$,

$f = 150\,kHz$, $\Delta V_{pp} = 0.1V$

$\Rightarrow T = 6.667\mu s$

$$D = \frac{5}{9} = 0.56$$

$$\Delta t = D \times T = (0.56)(6.667\mu)$$
$$= 3.73\mu s$$

OR $(0.8)(0.25)$ $= 0.2$

$$\Delta i_{pp} = \cancel{0.4025\ldots A}\left(\frac{I_{DC,LOAD}}{V_L}\right) = 0.2$$

$$I_{DC,\,I/p} = \frac{5 \times 0.8}{0.9 \times 9} = 0.4938 = 493.8m$$

$$\cancel{L = \frac{9 \times 3.73\mu}{0.}} \quad V_L = L\frac{\Delta i}{\Delta t}$$

$$V_L = 9 - 5 = 4$$

$$L = V_L \times \frac{\Delta t}{\Delta i_{pp}} = \frac{4(3.73\mu)}{(0.8/4)} = 74.6\mu H$$

$$C_{out} = 500\mu \times 0.8$$
$$= 400\mu F$$

$$ESR_{cout} = \frac{\Delta V_{pp}}{\Delta i_{pp}} = \frac{0.1}{0.2} = 0.5$$

$$P_{unreg} = P_{in} = 9 \times 0.49$$
$$= 4.41W$$

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EXAMPLE #5 (PROBLEM # 7.11) (Boost regulator)

$f = 100kHz$, $E_{unreg} = 5V_{DC}$, $V_{LOAD} = 9V_{,DC}$, $I_{DC,\,LOAD} = 0.8A$, DC,

$T = 1\times10^{-5}\,s$

$\Delta V_{PP} = 0.1V$

$\Delta t = 0.44\times1\times10^{-5}$

$= 4.4\mu s$

$D = 1 - \frac{5}{9} = 0.44$

$\Delta i_{PP} = 0.25\times I_{DC,\,I/P}$

$= 1.6\times0.25$

$= 0.4$

$I_{DC,\,I/P} = \frac{9\times0.8}{0.9\times5} = 1.6$

$L = \frac{E_{unreg}\times\Delta t}{\Delta i_{PP}} = \frac{5\times4.4\mu}{0.4}$

$L = 55\mu H$

$C_{out} = \frac{0.8\times4.4\mu}{\Delta V_{PP}} = 36\mu F$

$P_{unreg} = P_{in} = 5\times1.6$

$= 8W$

* Diode do not conduct in +ve.

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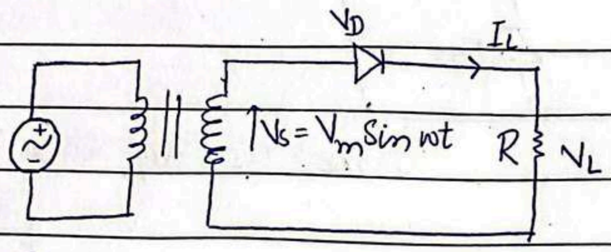
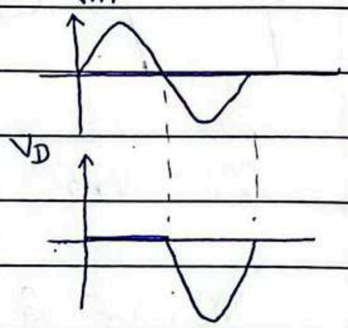
CHAPTER #10 (~~By~~ from power electronic converters by Haroon Rashid)

UNCONTROLLED RECTIFIERS:

Anode more +ve — diode ON.

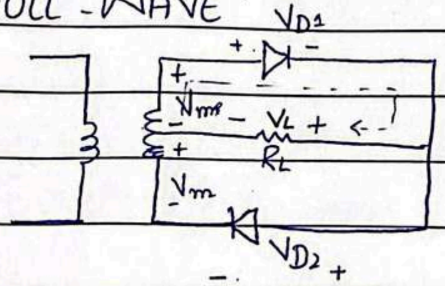
SINGLE - PHASE HALF WAVE RECTIFIER:

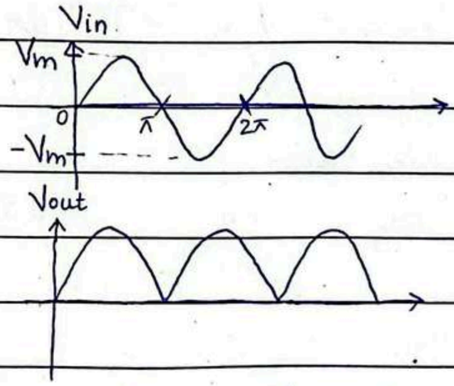


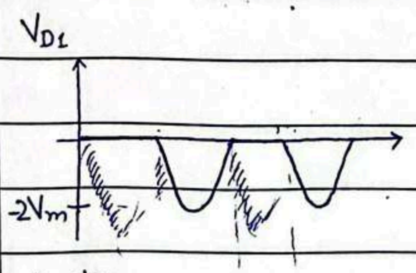


Peak inverse Voltage = PIV = V_m

FULL - WAVE

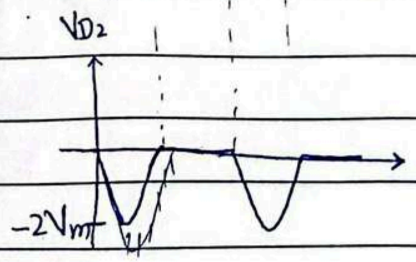






$V_m + V_m = 2V_m$ $\qquad V_{D1} + V_L$

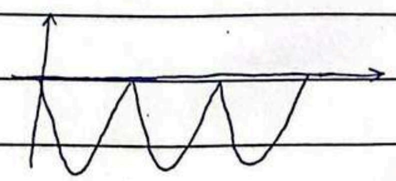
* $PIV = 2V_m$ $\quad (V_m + V_m)$



$V_{D1} + V_L + V_m = 0$

$V_D = V_{D1} + V_{D2}$

$\Rightarrow$



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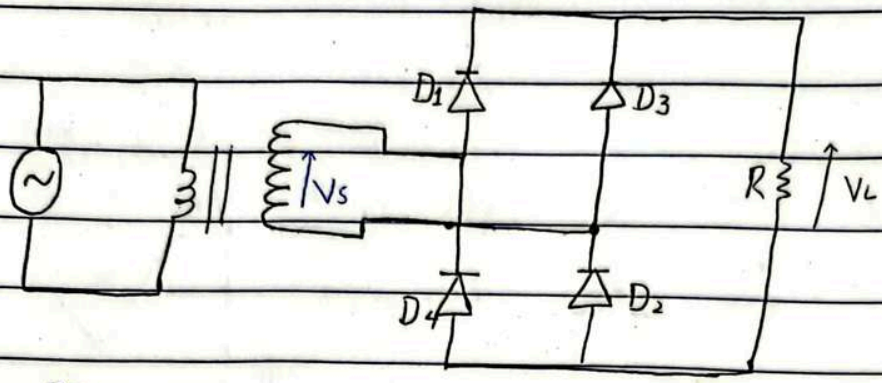
SINGLE-PHASE BRIDGE RECTIFIER



* PIV = V_m (same as half wave)

VOLTAGE RELATIONSHIPS

* $V_{DC} = \frac{1}{T}\int_0^T v(t)\,dt$

→ AVERAGE VALUE:

- Half wave:

$$V_{DC} = V_{avg} = \frac{V_m}{\pi} = 0.318 V_m$$

- full wave:

$$V_{DC} = V_{avg} = \frac{2V_m}{\pi} = 0.636 V_m.$$

→ RMS VALUE:

- Half wave:

$$V_{rms} = \frac{V_m}{2} = 0.5 V_m$$

- Full wave:

$$V_{rms} = \frac{V_m}{\sqrt{2}} = 0.707 V_m$$

CURRENT RELATIONSHIPS

- $I_{DC} = \dfrac{V_{DC}}{R}$ (Half wave = Full wave)
- $I_{RMS} = \dfrac{V_{rms}}{R}$ (Half wave = Full wave)

OUTPUT DC POWER:

- $P_{DC} = V_{DC} \times I_{DC}$

OUTPUT AC POWER

- $P_{AC} = V_{rms} \times I_{rms}$

RECTIFICATION RATIO (CONVERSION EFFICIENCY)

- $\sigma = \dfrac{P_{dc}}{P_{AC}} = \dfrac{V_{DC} \times I_{DC}}{V_{RMS} \times I_{RMS}}$

- *HALF WAVE:

$$\sigma = \frac{V_{DC} \times (V_{DC}/R)}{V_{RMS} \times (V_{rms}/R)} = \frac{V_{DC}^2}{V_{RMS}^2} = \frac{(V_m/\pi)^2}{(\cancel{2}V_m/\cancel{2}2)^2}$$

$$= \frac{\cancel{V_m^2}}{\pi^2} \times \frac{4}{\cancel{V_m^2}} = \frac{4}{\pi^2} = 0.405 = 40.5\%$$

- FULL WAVE:

$$\sigma = \frac{V_{DC}^2}{V_{RMS}^2} = \frac{(2V_m/\pi)^2}{(V_m/\sqrt{2})^2} = \frac{4\cancel{V_m^2}}{\pi^2} \times \frac{2}{\cancel{V_m^2}}$$

$$= \frac{8}{\pi^2} = 0.81 = 81\%$$

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FORM FACTOR (FF)

$$FF = \frac{V_{RMS}}{V_{DC}}$$

- HALF WAVE:

$$FF = \frac{V_m/2}{V_m/\pi} = \frac{\pi}{2} = 1.57$$

- FULL WAVE:

$$FF = \frac{V_m/\sqrt{2}}{2V_m/\pi} = 1.11$$

RIPPLE FACTOR:

* $$RF = \frac{V_{AC}}{V_{DC}} \quad (i)$$

$$\therefore V_{AC} = \sqrt{V_{RMS}^2 - V_{DC}^2}$$

$$\Rightarrow (i) \quad RF = \frac{\sqrt{V_{RMS}^2 - V_{DC}^2}}{\sqrt{V_{DC}^2}} = \sqrt{\frac{V_{RMS}^2}{V_{DC}^2} - 1}$$

$$= \sqrt{FF^2 - 1}$$

. TRANSFORMER UTILIZATION FACTOR:

$$TUF = \frac{P_{DC}}{V_s I_s} = \frac{V_{DC} \times I_{DC}}{V_s \times I_s}$$

* $$\therefore V_s = \frac{V_m}{\sqrt{2}} = 0.707\,V_m$$

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* $I_s = I_L$

$\Rightarrow$ * $I_{s\,(H.W)} = \frac{0.5\,V_m}{R}$, $I_{s\,(F.W)} = \frac{0.707\,V_m}{R}$

• HALF-WAVE:

$$TUF = \frac{(V_m/\pi)(V_{DC}/\cancel{R})}{(0.707\,V_m)(0.5\,V_m/\cancel{R})} \qquad V_{DC\,(H.W)} = V_m/\pi$$

$$= \frac{\cancel{V_m^2}/\pi^2}{0.3535\,\cancel{V_m^2}} = \frac{1}{0.3535\,\pi^2}$$

$$TUF = 0.286$$

• FULL-WAVE:

$$TUF = \frac{P_{DC}}{2 \times V_s \times I_s} = \frac{(2V_m/\pi)(2V_m/\pi R)}{2 \times (V_m/\sqrt{2})(0.5\,V_m/R)}$$

for center tapped; full wave $V_s I_s$ is double of half wave. ~~rectifiers oper~~ 2 half wave rectifiers are operating together.

$$= \frac{4V_m^2/\pi^2 \cancel{R}}{2 \times (0.5\,V_m^2)/\sqrt{2}\,\cancel{R}}$$

$$= \frac{4}{\pi^2} \times \frac{\sqrt{2}}{0.5} \times \frac{1}{2} = 0.572$$

• BRIDGE RECTIFIER:

$$TUF = \frac{P_{DC}}{V_s I_s} = \frac{V_{DC} \times I_{DC}}{V_s \times I_s}$$

$$= \frac{(2\cancel{V_m}/\pi)(2\cancel{V_m}/\pi R)}{(0.707\cancel{V_m})(0.707\cancel{V_m}/\cancel{R})} = \frac{4/\pi^2}{(0.707)^2}$$

$$TUF = 0.81$$

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POWER FACTOR:

$$PF = \frac{P_{AC}}{V_S \times I_S}$$

CREST FACTOR: (measure of peak I/P current compared to RMS of I_L).

$$CF = \frac{I_{S\,(PEAK)}}{I_S}$$

EXAMPLE # 1

Full wave transformer - center tapped:

i) Efficiency = $\sigma = \dfrac{P_{DC}}{P_{AC}} = \dfrac{V_{DC} \times I_{DC}}{V_{RMS} \times I_{RMS}}$

$$= \frac{(2V_m/\pi)(2V_m/\pi R)}{(0.707\,V_m)(0.707\,V_m/R)}$$

$$= \frac{4/\pi^2}{(0.707)^2}$$

$$\sigma = 0.81$$

ii) Form Factor $= FF = \dfrac{V_{RMS}}{V_{DC}} = \dfrac{V_m/\sqrt{2}}{2V_m/\pi}$

$$= \frac{V_m}{\sqrt{2}} \times \frac{\pi}{2V_m}$$

$$FF = 1.11$$

iii) Ripple factor $= RF = \sqrt{FF^2 - 1}$

$$= \sqrt{(1.11)^2 - 1}$$

$$= 0.48$$

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iv) $TUF = 0.572$

v) $PIV_{D1} = 2V_m$ (PIV is same on both dides $\Rightarrow 2V_m$)

$\therefore PIV_{F.W} = 2V_m$

vi) $CF = \dfrac{I_{S(PEAK)}}{I_S}$ $RMS = \dfrac{PEAK}{\sqrt{2}}$

$I_{S(PEAK)} = \dfrac{\sqrt{2} \times 0.707 V_m}{R}$ $\therefore I_{S_{F.W}} = \dfrac{0.707\,V_m}{R}$

$I_{S(PEAK)} = \dfrac{0.999 V_m}{R}$

$\Rightarrow CF = \dfrac{0.999 \cancel{V_m}/\cancel{R}}{0.707 \cancel{V_m}/\cancel{R}} = 1.414$

vii) $PF = \dfrac{P_{AC}}{V_S \times I_S} = \dfrac{V_{RMS} \times I_{RMS}}{V_S \times I_S}$

$= \dfrac{(\cancel{V_m}/\sqrt{2})(\cancel{V_m}/\sqrt{2}\cancel{R})}{(0.707 \cancel{V_m})(0.707 \cancel{V_m}/\cancel{R})}$

$= \dfrac{1/(\sqrt{2})^2}{(0.707)^2}$

$PF = 1$

23/4 EXAMPLE # 2

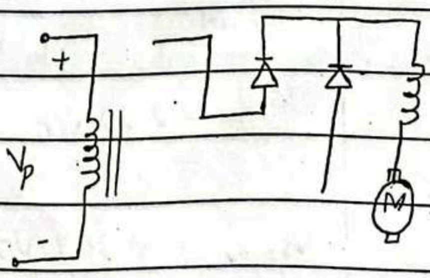
Single phase bridge rectifier

$TUF = 1$, $I_{S1} = 0.9\, I_a$

$I_S = I_a$, $\phi = 0^\circ \Rightarrow \cos 0^\circ = 1$

$HF = ?$, $PF = ?$

$HF = THD$



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* $HF = \left(\frac{I_s^2 - I_{s1}^2}{I_{s1}^2}\right)^{1/2} = \left[\left(\frac{I_s}{I_{s1}}\right)^2 - 1\right]^{1/2}$

$= \sqrt{\left(\frac{\cancel{I_a}}{0.9\cancel{I_a}}\right)^2 - 1} = 0.484$

* $PF = \frac{I_{s1}}{I_s} \cos\phi = \frac{0.9 I_a}{I_a} (\cos 0^\circ)$

$= 0.9$ (lagging)

* → inductive load — lagging
→ capacitive load — leading

DESIGN CONSIDERATIONS

If, $V_{DC} = 10V$

For Half wave:

i) V_s: $V_{DC} = 0.318 V_m$, $V_{RMS} = 0.5 V_m$] O/P $\Rightarrow V_m = V_{DC}/0.318$

* $V_{S, \text{I/P secondary voltage}} = V_S = 0.707 V_m$

$V_{S_{H.W}} = 0.707 \frac{V_{DC}}{0.318}$

$= 2.22 V_{DC}$

$V_{S_{F.W}} = 0.707 \frac{V_{DC}}{0.636}$ $\qquad V_{DC, F.W} = 0.636 V_m$

$= 1.11$

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ii) PIV: ↗ Peak repetitive reverse voltage

→ Half wave: $V_{RRM} = V_m = \underbrace{\frac{V_{DC}}{0.318}}_{V_m} = 3.14\ V_{DC}$

$\therefore PIV_{H.W} = V_m$

→ Full wave

- Center tapped $= V_{RRM} = 2V_m = \underbrace{\frac{V_{DC}}{0.636}}_{V_m} \times 2 = 2 \times 1.57\ V_{DC} = 3.14\ V_{DC}$

$PIV_{F.W} = 2V_m$ (center tapped)

- Bridge rectifier $= V_{RRM} = V_m = \frac{V_{DC}}{0.636} = 1.57\ V_{DC}$

PIV (bridge rectifier) $= V_m$

iii) Peak repetitive forward current $= I_{FRM} = V_m/R$

→ Half wave: $I_{DC} = V_{DC}/R \Rightarrow I_{DC} \times R$

$$I_{FRM} = \frac{V_{DC}}{0.318} = \frac{I_{DC} \times \cancel{R}}{0.318\cancel{R}} \qquad V_m =$$

$$= 3.14\ I_{DC}$$

→ Full wave:

$$I_{FRM} = \frac{V_{DC}}{0.636\ R} = 1.57\ I_{DC}$$

THREE-PHASE UNCONTROLLED RECTIFIER:

- Single phase — used for low to medium power application
- For output higher than 15kW, three phase rectifiers are used.
- TYPES:

→ Star rectifier

→ Bridge rectifier

* Full wave - conduction angle = 360

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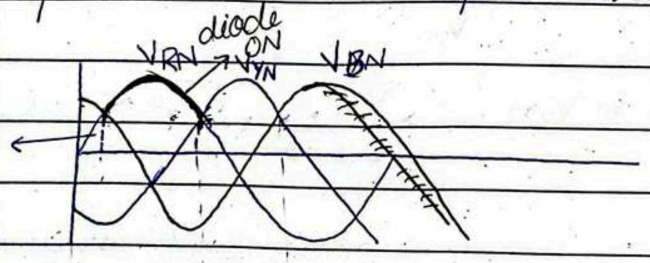
→ STAR RECTIFIERS (Half wave)

- Diode - ON: when voltage on that phase is greater than the other two.
- Diode conduction angle: $2\pi/3$ instead of π.

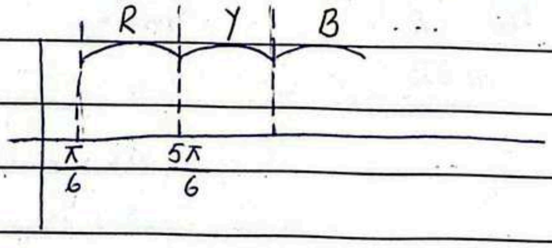
→ 1 phase → 1 diode ⇒ half wave

→ 2 phase → 2 diode ⇒ full wave.



consider ideal diode

V_{RN} - phase to neutral



$V_m \sin\omega t \, d\omega t$

$= V_m \sin\theta \, d\theta$

$$T_{ON} = \frac{5\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{3} \text{ rad}$$

$$\text{Conduction angle} = 120^\circ \text{ OR } \frac{2\pi}{3} \text{ rad}$$

$$V_{DC} = \frac{1}{2\pi/3} \int_{\pi/6}^{\frac{5\pi}{6}} V_m \sin\theta \, d\theta$$

$T_{ON} \leftarrow$

$$= \frac{3}{2\pi} \int_{\pi/6}^{5\pi/6} V_m \sin\theta \, d\theta$$

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$V_{DC} = 0.8269\, V_m$

$\approx 0.827 V_m$

$$V_{RMS} = \sqrt{\frac{3}{2\pi}\int_{\pi/6}^{5\pi/6} V_m^2 \sin^2\theta \, d\theta}$$ ~~RMS = PEAK/√2~~

$V_{RMS} = 0.84\, V_m$

I_s works one time for each winding

$$I_s = \sqrt{\frac{1}{2\pi}\int_{\pi/6}^{5\pi/6} I_m^2 \sin^2\theta \, d\theta}$$

$I_s = 0.485\, I_m$

$I_s = 0.485 \dfrac{V_m}{R}$

EXAMPLE #3

A three phase star rectifier :

i) Efficiency $= \sigma = \dfrac{V_{DC} \times I_{DC}}{V_{RMS} \times I_{RMS}}$

$$= \frac{0.827 V_m \times \overset{0.827}{\cancel{0.485}} V_m/R}{0.84 V_m \times V_m 0.84/R}$$ $I_{RMS} = V_{RMS}/R$

$= \cancel{0.568}\ 0.969 = 96.9\%$

ii) $FF = \dfrac{V_{RMS}}{V_{DC}} = 1.015$

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iii) $RF = \sqrt{FF^2 - 1} = \sqrt{(1.01)^2 - 1}$

~~0.1238~~ ~~0.1414~~ $= 0.141$

iv) $TUF = \frac{P_{DC}}{3 \times V_s \times I_s} = \frac{V_{DC} \times I_{DC}}{3 \times V_s \times I_s}$

for 3-phase ←

$\therefore V_s = 0.707 V_m$

$I_s = 0.485 I_m$

$TUF = (0.827)^2 / (3 \times 0.707 \times 0.485)$

$TUF = 0.66 < 1$ * (always less than 1)

←

* $V_{ab} = V_a - V_b$

Phase to phase voltage OR line-to-line voltage (line voltage) V_L

Phase-to-neutral voltage (Phase voltage V_P) OR line-to-neutral voltage.

* $V_L = \sqrt{3} V_P$ $(1\angle 0 - 1\angle 120 = \sqrt{3})$

⇒ $PIV = \sqrt{3} V_m$

line rating

→ v) $PF = \frac{P_{AC}}{3 \times V_s \times I_s} = \frac{0.84^2}{3 \times 0.707 \times 0.485}$

$= 0.685$

vi) $PIV = V_m$

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vii) $I_{DC} = 30A$ (given) , $V_{DC} = 140V$ (given)

$I_{Diode(avg)} = \frac{30}{3} = 10$ (i)

$I_{DC} = 0.827 I_m$

$I_{diode(avg)} = 0.275\cancel{87} I_m$ (ii) : $0.827/3$

Compare eq. (i) & (ii)

$10 = 0.2757 I_m$

$I_{m(Peak)} = 36.27A$

30/4 THREE PHASE BRIDGE RECTIFIERS:

- Conduction angle $= \frac{2\pi}{3}$
- highest possible TUF

→ No. of diode shows how it will ON.

→ At a time, 2 diodes will be ON.

- Line voltage $= \sqrt{3} V_m$

$V_P = \sqrt{3} V_m = \cancel{1.73 V_m}$

* $V_{DC} = V_{AVG} = \frac{6}{2\pi} \int_{\pi/3}^{2\pi/3} \sqrt{3} V_m \sin\theta d\theta$: (6 waves)

$= 1.6539$

$= 1.65\cancel{4} V_m$

* $V_{RMS} = \sqrt{\frac{6}{2\pi} \int_{\pi/3}^{2\pi/3} (\sqrt{3} V_m \sin\theta)^2 d\theta}$

$= 1.655 V_m$.

* $I_{s(AVG)} = \frac{4}{2\pi}\int_{\pi/3}^{\frac{2\pi}{3}} I_m \sin\theta\, d\theta$

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- $I_s = \sqrt{\frac{4}{2\pi}\int_{\pi/3}^{2\pi/3} I_m^2 \sin^2\theta\, d\theta}$ (in transformer: current flows only if related diodes are ON)

$I_s = 0.78\ I_m$ (each winding - transformer secondary current)

- $I_m = \frac{\sqrt{3}\,V_m}{R}$

- Diode current $= \frac{\sqrt{3}\,V_m}{R}$ $\quad I_{D_{RMS}} = \sqrt{\frac{2}{2\pi}\int_{\pi/3}^{2\pi/3} I_m^2 \sin^2\theta\, d\theta}$

$= 0.552\, I_m$

- $TUF = \frac{(1.654)^2}{3 I_s V_s}$

$= \frac{(1.654\, V_m)(1.654\, V_m/R)}{3\times 0.707\, V_m \times 0.78 \times \sqrt{3}\, V_m/R}$

$= 0.954$

EXAMPLE #4 (from Power electronics 4th Edition by Haroon Rashid)

$V_{DC} = 280.7\ V$ (i) & $V_{DC} = 1.654\, V_m$ (ii)

from (i) & (ii) $\Rightarrow V_m = 169.71\, V$ $\qquad V_m = \frac{280.7}{1.654}$

$P.I.V. = \sqrt{3}\, V_m$

$= 293.94$

* if not given, consider it as RMS value.

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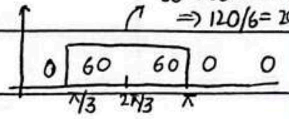
EXAMPLE #5 (Line to neutral voltage)

$I_{DC} = 60A$, $V_{supply} = 120$

$60+60=120 \Rightarrow 120/6 = 20$



→ $I_{D(avg)} = \frac{1}{2\pi}\int_{\pi/3}^{\pi}(I_{DC}\, d\theta) = 20A$

→ $I_{D(RMS)} = \sqrt{\frac{1}{2\pi}\int_{\pi/3}^{\pi} 60^2\, d\theta}$

$= 34.64$ OR $\frac{I_{DC}}{\sqrt{3}} = \frac{60}{\sqrt{3}} = 34.64$

→ $PIV = \sqrt{3}\ V_m$

$= \sqrt{3} \times 120_{RMS}$ ∵ $V_{RMS} = \frac{V_{peak}}{\sqrt{2}}$

$= \sqrt{3} \times \sqrt{2} \times 120$

$= 293.93$

EXAMPLE #6

a) $I_{RMS} = ?$ $t_1 = 100\mu s$, $t_2 = 350\mu s$, $t_3 = 500\mu s$,

$f = 250Hz$, $f_s = 5kHz$, $I_m = 450A$, $I_a = 150A$

$I_{(AVG)} = \frac{1}{T}\int_0^{t_1} I_m \sin w_o t + \frac{1}{T}\int_{t_2}^{t_3} i_2\, dt$, $i_2 = I_a$

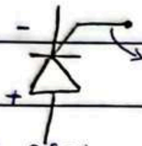
$\frac{1}{1/f} = 250$, $2\pi\, 5000\, t$ ∵ $w_o = 2\pi f_s = (2\pi\, 5000)$

$= 250\left[\int_0^{100\mu} 450 \sin(2\pi\, 5000\, t)\, dt + \int_{350\mu}^{500\mu} 150\, dt\right]$

$= 12.79$

$I_{RMS} = 58.09$

THYRISTORS (~~un~~controlled rectifiers)



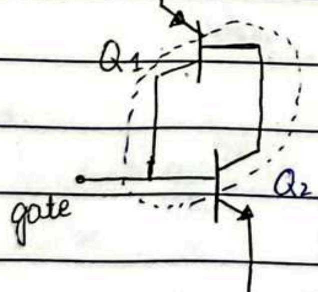
(it will not work until trigger is applied at gate)

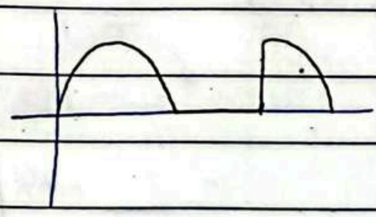
(will also work if voltage is increased - No gate pulse applied)

- Delivers 1000's of W to the load of 98% efficiency.

→ SCR (Silicon controlled rectifiers)

- Circuits that protect thyristors and line are called Snubbers.
- Q_2 keeps Q_1 ON, Q_1 keeps Q_2 ON.
- Thyristors – 4 layers ⇒ PnPn
- Inherently they are always OFF.
- To Stop working thyristors - reduce its current.

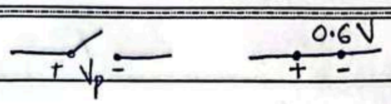




5/5 SILICON CONTROLLED RECTIFIER

- Current above $\underbrace{\text{holding current}}_{I_H}$ ⇒ ~~SR~~ SCR - ON

- $V_{TM(min)} = 0.6V$



- I_{TSM} : Surge current
  Thousands of Watts delivers to load.

- $I_{RMS} = I_P/2$
- V_{gt} :
  0.7V to forward-bias the gate ~~collectors~~ junction.

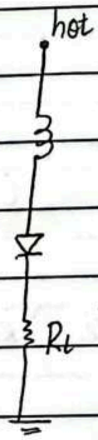
SNUBBING

- Thyristors turn on in 10μ to 100μs.
- Voltage across them ↓ (line peak of 100's to 2V)
- Current increases from 0 to tens to hundreds Amperes

→ to control di/dt place an inductor in series with the thyristors.
in snubbers. (CRITICAL RATE OF RISE ~~IN~~ OF CURRENT)

$$* \quad L > \frac{V_{P,LINE\,(Peak)}}{(di/dt)_{\text{spec.}}}$$

- EXAMPLE #1



$V_{LINE} = 220V_{RMS}$, $f_{LINE} = 50Hz$
$R_L = 80\Omega$, $di/dt = 50A/\mu s$.

a) $$L > \frac{220\sqrt{2}}{50/\mu} = \cancel{4.4\mu H}$$

$$L > 6.22\ \mu H$$

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b) $Z_L = j\omega L$ (impedance of inductor)

$= (2\pi(50))(6.2\mu)\angle 90^\circ \rightarrow +j$

$= 1.95m \angle 90^\circ$

$= 0.00195 \angle 90^\circ$

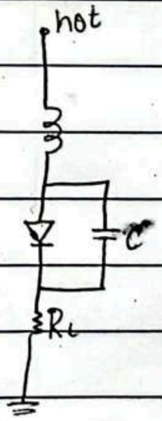
CRITICAL RATE OF RISE OF VOLTAGE:

- Thyristor may turn on due to noise (w/o gate pulse) because of parasitic capacitance b/w main terminal & gate.

→ $V_C = V_{P,noise}\left(1 - e^{-t/(R_L \times C_{snub})}\right)$

→ $\tau = 1/RC \Rightarrow$ natural response $= V_0 e^{-t/\tau} = V_0 e^{-t/RC}$

→ $\dfrac{dV_{C(snub)}}{dt} = \dfrac{V_{P,noise}}{R_L \times C_{snub}} e^{-t/(R_L \times C_{snub})}$



→ Worst case, t = 0 capacitor charges quickly.

$$\left.\frac{dV_{C(snub)}}{dt}\right|_{\text{worst case}, t=0} = \frac{V_{P,noise}}{R_L \times C_{snub}}$$

→ $i = C\dfrac{dV_C}{dt}$

* • Use V_{RMS} as $V_{P,noise}$

$$C_{snub} > \frac{V_{DRM}}{R_{LOAD} \times (dV/dt)_{specs}}$$

$(dV/dt)_{specs}$ → data by manufacturer

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EXAMPLE #2

$V_{LINE} = 220 V_{RMS}$, $f_{LINE} = 50Hz$, $V_{DRM} = 400 V_{(PEAK)}$

$R_L = 80\Omega$, $dv/dt = 75/\mu V$

a) $$C > \frac{V_{RM}}{R_L \times (dv/dt)} = \frac{400}{80(75/\mu)}$$

$$C > 66.7 nF = 0.067 \mu F$$

$$C \simeq 0.1 \mu F$$

b) $$i = C\frac{dv}{dt} = (\cancel{0}.067\mu)\left(\frac{75}{\mu}\right)$$

$$Z_C = \frac{1}{j\omega C}$$ *$X_C = 1/\omega C$ (w/o j)

$$= \frac{1}{(2\pi \times 50)(0.1\mu)} \angle -90^\circ$$

$$= 31830.9 \angle -90^\circ$$

$$= 32k \angle -90^\circ$$

$$I_{Leakage} \overset{*}{=} I_{LOAD}\ I_{RMS} = \frac{V_{RMS}}{R_{LOAD} + Z_C} = \frac{220}{(80) + (32k \angle -90^\circ)}$$ * $I_L = I_{RMS}$

$$= 6.87_m \angle 89.85$$

$$= 6.9 \angle 90^\circ mA$$

$$V_{LOAD} = (6.9 \angle 90m)(80)$$

$$= 0.5 V_{RMS} \angle 90^\circ$$

$$P_{LOAD} = I_{LOAD} \times V_{LOAD}$$

$$= 3m W$$

Date:

LEAKAGE CURRENT:

- Current through capacitor is leakage current.

- $$I_{discharge} = \frac{V_{P,LINE}}{R_{snubbing}}$$

- Total current at anode becomes:

$$I_{SCR} = I_{P,LOAD} + I_{Discharge} < I_{TSM}$$

$$I_{TSM} > I_{P,LOAD} + I_{discharge}$$

$$\Rightarrow R_{snubbing} > \frac{V_{P,LINE}}{I_{TSM} - I_{P,LOAD}}$$

EXAMPLE #3

$V_{Line} = 220\,V_{RMS}$, $f_{LINE} = 50\,Hz$;

$R_{LOAD} = 80\Omega$, $I_{TSM} = 46\,A_P$.

$$R_{snubbing} > \frac{\overbrace{220\sqrt{2}}^{V_{LINE}}}{46 - \left(\frac{220\sqrt{2}}{\underbrace{80}_{R_{LOAD}}}\right)}$$

$$= 7.38\Omega$$